Steve (Kai) Fan

Limited Term Assistant Professor · University of Georgia

Contact

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Research Interests

Analytic number theory: the distribution of primes, zeta and L-functions, arithmetic functions, sieve methods, with applications to arithmetic statistics

Additive combinatorics: additive structure of arithmetic sets, including Sárközy-type problems and completeness of sets and sequences

Education

Nov 2023 **Ph.D. in Mathematics, Dartmouth College**
Advisor: Carl Pomerance.
Dissertation: *Rough Numbers and Variations on the Erdős–Kac Theorem*.

Feb 2020 **M.A. in Mathematics, Dartmouth College**

Jun 2015 **B.S. in Mathematics, Southeast University**

Academic Appointments

Jan 2025–present **Limited Term Assistant Professor**
Department of Mathematics, University of Georgia.
Postdoctoral mentor: Paul Pollack.

Jan 2024–Nov 2024 **Postdoctoral Fellow**
Max Planck Institute for Mathematics, Bonn, Germany.

Publications and preprints

[1] **On the van der Corput property for cubes**
Steve Fan, Andrew Lott, Ákos Magyar, Máté Matolcsi, and Imre Z. Ruzsa
Preprint (2026). arXiv:2611.05267.

Publications and preprints (continued)

[2] **The optimal van der Corput property for squares**
Steve Fan and Andrew Lott
Preprint (2026). arXiv:2610.07765v2.

[3] **Rank stability of elliptic curves over $\mathbb{F}_q(t)$ in residue classes**
Steve Fan and Sun Woo Park
Preprint (2026). arXiv:2609.33792.

[4] **Strongly complete sets and a conjecture of Erdős**
Steve Fan
Preprint (2026). arXiv:2607.14071.

[5] **The van der Corput property for sums of two squares**
Steve Fan and Andrew Lott
Preprint (2026). arXiv:2606.29185.

[6] **On the asymptotic density of the ordered pairs (a, b) of positive integers such that $\gcd(ab, a+b) = \gcd(a, b)$**
Steve Fan and László Tóth
Preprint (2026). arXiv:2606.20057v3.

[7] **Extensions of the Furstenberg–Sárközy theorem via the arithmetic level-d inequality**
Carlo Francisco E. Adajar, Rishika Agrawal, Mukul Rai Choudhuri, Chian Yeong Chuah, Steve Fan, Swaroop Hegde, Andrew Lott, Krishnamohan Nandakumar, and Nagendar Reddy Ponagandla
Preprint (2026). arXiv:2605.16216.

[8] **On a super telescoping sum representing binomial coefficients**
Steve Fan
Rocky Mountain J. Math. **56**(1) (2026), 137–143.

[9] **A family of analogues to the Robin criterion**
Steve Fan, Mits Kobayashi, and Grant Molnar
Ramanujan J. **70** (2026), Article 40, 61 pp.

[10] **Sárközy's theorem in $\mathbb{F}_q[t]$ via the van der Corput property**
Steve Fan and Andrew Lott
Finite Fields Appl. **116** (2026), 40 pp.

[11] **The maximal order of the shifted-prime divisor function**
Steve Fan and Paul Pollack
Integers **26A** (2026), A9, 14 pp.

[12] **The Hardy–Ramanujan inequality for sifted sets and its applications**
Steve Fan
Preprint (2025). arXiv:2508.06005.

[13] **Counting primes with a given primitive root, uniformly**
Steve Fan and Paul Pollack
Mathematika **71**(4) (2025), e70055, 30 pp.

Publications and preprints (continued)

[14] **Extremal elasticity of quadratic orders**
Steve Fan and Paul Pollack
In *The ideal theory and arithmetic of rings, monoids, and semigroups*, Scott T. Chapman (ed.), Contemp. Math. 836, Amer. Math. Soc., Providence, RI, 2026, pp. 35–53.

[15] **The typical elasticity of a quadratic order**
Steve Fan and Paul Pollack
arXiv:2411.09063 (2024). Revised manuscript under review at *Math. Z.*

[16] **The shifted prime-divisor function over shifted primes**
Steve Fan
Ramanujan J. **66** (2025), Article 44, 46 pp.

[17] **Shifted-prime divisors**
Steve Fan and Carl Pomerance
arXiv:2401.10427 (2024). To appear in a Springer volume dedicated to Helmut Maier on his 70th birthday, edited by J. Friedlander, C. Pomerance, and M. Rassias.

[18] **Weighted Erdős–Kac theorems via computing moments**
Steve Fan
Acta Arith. **217**(2) (2025), 99–158.

[19] **Numerically explicit estimates for the distribution of rough numbers**
Steve Fan
J. Number Theory **260** (2024), 120–150.

[20] **An inequality related to the sieve of Eratosthenes**
Steve Fan and Carl Pomerance
J. Number Theory **254** (2024), 169–183.

[21] **An inequality for the distribution of numbers free of small prime factors**
Steve Fan
Integers **22** (2022), A26, 12 pp.

[22] **The second largest Balaban index (sum-Balaban index) of unicyclic graphs**
Wei Fang, Yubin Gao, Kai Fan, and Zhongshan Li
J. Math. Res. Appl. **37**(4) (2017), 391–403.

[23] **A finite difference scheme for semilinear space-fractional diffusion equations with time delay**
Zhaopeng Hao, Kai Fan, Wanrong Cao, and Zhizhong Sun
Appl. Math. Comput. **275** (2016), 238–254.

Recent Talks

Sept 2026	**The van der Corput property for sums of two squares** Number Theory Seminar, Duke University.
May 2025	**Counting shifted-prime divisors** Plenary talk, Integers Conference 2025.
Mar 2025	**The typical elasticity of a quadratic order** Number Theory/Algebraic Geometry Seminar, University of Georgia.

Recent Talks (continued)

Nov 2024	**The typical elasticity of a quadratic order** The Fouvry Seminar, University of Paris-Saclay, Orsay, France.
Oct 2024	**The typical elasticity of a quadratic order** Number Theory Lunch Seminar, Max Planck Institute for Mathematics.
May 2024	**Counting shifted-prime divisors** Intercity Number Theory Seminar, Utrecht University.

Teaching

University of Georgia	Instructor of record
Fall 2026	MATH 4400/6400: Number Theory
Spring 2026	MATH 2700: Differential Equations
Fall 2025	MATH 2250: Calculus I Two sections
Spring 2025	MATH 1113: Precalculus Two sections
University of Georgia	Directed Reading Program Mentor
Spring 2026	Roth's theorem on three-term arithmetic progressions Mentee: Rishab Kalyan Subramaniyan
Dartmouth College	Instructor of record
Fall 2022	MATH 11: Accelerated Multivariable Calculus
Fall 2021	MATH 1: Introduction to Calculus
Summer 2020	Math Camp: Exploring Mathematics
Dartmouth College	Teaching assistant
Spring 2020	MATH 13: Multivariable Calculus
Fall 2019	MATH 23: Differential Equations
Winter 2019	MATH 3: Calculus
Fall 2018	MATH 3: Calculus